Akachidike David Nweze

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EDUCATION

New Jersey Institute of Technology Newark, NJ
Bachelor of Science, Data Science. Dean's List, NJIT Merit Scholarship. GPA: 3.7/4.0. May 2029
Relevant Coursework: Data Structures, Linear Algebra, Discrete Mathematics, Calculus, Program Development

EXPERIENCE

Undergraduate Research Assistant, James Geller Medical Ontology Research Lab Sep 2026 – Present
New Jersey Institute of Technology Newark, NJ

- Conduct research in biomedical informatics and ontology engineering, contributing to AI-driven projects involving 100,000+ medical concepts from large healthcare knowledge bases such as UMLS and SNOMED CT.
- Analyze and curate 10,000+ ontology relationships to improve the consistency and quality of biomedical knowledge graphs.
- Build data-processing pipelines in Python and SQL to clean, transform, and evaluate large structured datasets.
- Collaborate with a multidisciplinary team of 5+ faculty members, graduate researchers, and undergraduate students on medical AI and semantic web research projects.

Peer Tutor Sep 2026 – Present
New Jersey Institute of Technology Newark, NJ

- Tutor 25+ undergraduates one-on-one and in small groups in Calculus I/II and introductory Python programming.
- Conduct 8-12 tutoring sessions per week, helping students strengthen problem-solving, programming, and analytical skills.

Information Technology Intern May 2025 – Aug 2025
Global Process and Pipeline Services Port Harcourt, Nigeria

- Provided technical support to 50+ employees across engineering, operations, and administrative departments, resolving hardware, software, and network issues.
- Analyzed IT asset and support data in Excel, tracking 100+ devices and identifying ways to improve equipment utilization.

PROJECTS

Email Uplift Modeling: Who Should Get the Email? Nov 2026
Python, uplift models, causal forest | *View Project & Findings*

- Built uplift models (S-, T-, X-learner, causal forest) to estimate which customers respond to a marketing email, using a randomized experiment with 64,000 customers.
- Validated targeting policies with cross-fitting; a profit gain seen on one test set did not replicate, so emailing everyone was the stronger policy for the women's email.
- Found the men's email performed about equally across customer segments, and sending it to everyone had the highest estimated profit.

NBA Load Management Analysis Oct 2026
Python, pandas, statsmodels, Matplotlib | *View Project & Findings*

- Tested whether the NBA's 2023 rule requiring star players to play more games worked, using 160,000+ player-games.
- Found stars sat out fewer back-to-back games after the rule, but did not play more games over a full season.
- Created a data-driven defensive rating that ranks All-Defensive Team players above others 89% of the time.

NBA Rest Analytics Pipeline Oct 2026
SQL, dbt, DuckDB, Tableau, Python | *View Project & Findings*

- Built an automated, quality-tested SQL pipeline that turns raw NBA game logs into clean, analysis-ready data.
- Found teams playing their second game in two nights win about 6 fewer games per 100, as much as home-court advantage.
- Published an interactive Tableau dashboard that lets anyone explore the results by team and season.

ACTIVITIES & LEADERSHIP

Student Activities Council (SAC), NJIT Aug 2026 – Present
Treasurer

- Track and manage $110,000+ in organizational spending across 20+ campus events, monitoring expenses against allocated budgets to ensure accurate financial reporting and responsible decision-making.

SKILLS

Programming Languages: Python, Java, SQL
Data and ML: Pandas, NumPy, Scikit-learn, statsmodels, Matplotlib, Seaborn, Data Analysis, Machine Learning
Causal Inference: Uplift Modeling, Causal Forests, Cross-Fitting, Treatment-Effect Estimation
Statistics: Regression, Difference-in-Differences, Hypothesis Testing
Databases: PostgreSQL, MySQL, DuckDB, Relational Database Design
Tools: Tableau, dbt, Git, GitHub, Jupyter Notebook